

OCRFix-RU

Hybrid Word+Character N-gram for OCR Error Correction

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1) Why This Matters

OCR noise breaks token semantics and downstream NLP quality.

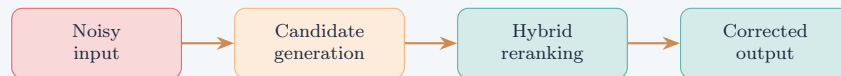
- Visual confusions in Cyrillic text distort words and context.
- We need a lightweight, interpretable, robust correction method.

Examples: **к0т** → **кот**, **т3кст** → **текст**

2) Minimal Experimental Setup

- **Dataset:** synthetic OCR-noisy Russian text.
- **Baseline:** vanilla word n-gram corrector.
- **Method:** hybrid word+char n-gram with OCR-aware reranking.
- **Metrics:** Accuracy, WER, CER.

3) Correction Pipeline (Core Idea)



$$S(w) = \alpha \log P_{word}(w | c) + (1 - \alpha) \log P_{char}(w) - \lambda \cdot ED$$

4) Hybrid Components

Word LM

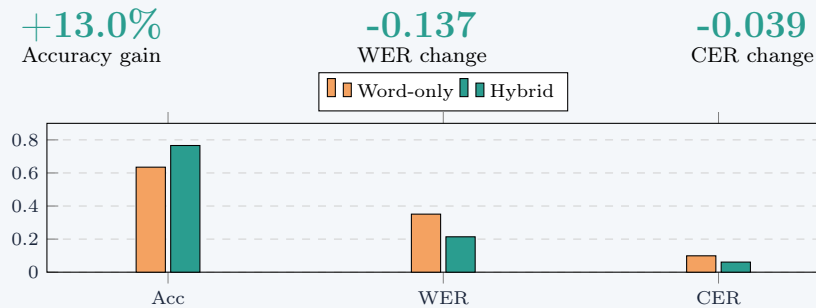
context score
word-form score
confusion match

Char LM

OCR Channel

Vanilla baseline uses only Word LM; hybrid combines all three signals.

5) Main Results (Headline)



6) Error Patterns (Qualitative)

к0т → **кот**
т3кст → **текст**
пара → **пара**
модель → often unresolved (OOV)

- Best delta=0.020 at alpha=1.00, noise=0.30.
- Best ablation delta: **0.020**.
- Gains are stronger at higher synthetic noise levels.

7) Big Takeaway

Hybrid model significantly improves OCR correction under noisy conditions.

- Hybrid outperforms the vanilla word n-gram baseline in the main setup.
- Hybrid improves both WER and CER on the main setup.